

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

Spatial overlap and mean class contrast need not imply target-specific region responses. We audit CCI’s maximum-target-drop region in frozen CLIP, using separate prompts for selection and evaluation. Across COCO and VOC with two checkpoints, 41.08–64.78% of locality/mean-foil screen-passers fail a worst-foil criterion. With original screening memberships retained, excluding annotated-present foils leaves 39.69–63.64% failing. Exclusion reduces failure beyond exact cardinality-matched random removal by 0.86–0.92 percentage points on COCO; VOC intervals include zero. Positive-target filtering retains high failure. Among screened failures, unrestricted eight-region oracles permit sign repair in 6.25–7.84% on COCO and 27.40–31.15% on VOC. A target-drop tolerance of $\epsilon = .02$ reduces feasible repair to 0.16–0.98%; relaxing it increases opportunity with setting-dependent target and locality changes.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

CLIP [1] region explanations use intervention-induced image–text score changes. Cluster-based Concept Importance (CCI) [2] groups patches and measures their contribution; we audit its maximum-target-drop region, CCI-top1. A positive mean-foil contrast can average away a strong competitor: masking a region may reduce the target score more than the average non-target score, yet reduce one non-target score even more. Worst-foil specificity tests this strongest competitor, complementing spatial overlap and mean contrast.

Wang and Wang [3] analyze mean, max, and weighted class contrast. Their max contrast chooses the non-target class with the highest original-image logit; our worst foil maximizes the score drop under a region intervention and can vary with the region. CCI-top1 isolates the region selected for maximum target drop, enabling a direct audit of its cross-class specificity.

COCOA [4] explains corpus similarity relative to foil samples; CASE [5] tests class sensitivity and constructs contrastive saliency. Our contribution combines fixed-population auditing, exact cardinality-matched foil controls, and candidate/budget decomposition to quantify worst-foil failure and decompose

constraints on sign repair. Related CLIP attribution [6, 7, 8], class contrasts [9], and caption foils [10] provide complementary perspectives.

2 AUDIT AND REPAIR OPPORTUNITY

Figure 1 links candidate construction from frozen patch features (a), one-prompt selection (b), and three-prompt evaluation of the fixed region (c).

Let x be an image, t its target class, and c any dataset class. A candidate $R \in \mathcal{R}(x)$ is a set of image patches; $\mathcal{R}(x)$ contains $K = 8$ candidates. For checkpoint m , the pre-softmax score is $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$. Here z_m and e_m are unit-norm image and prompted-text embeddings, $\langle \cdot, \cdot \rangle$ is their inner product, and τ_m is CLIP’s learned positive logit scale. The intervention $\mathcal{I}(x, R)$ masks attention to patches in R . The resulting score drop is

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

Positive $d_R(c)$ means a score reduction. Superscript sel denotes the prompt “a photo of a {category}”. CCI-top1 selects

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Held-out margins and screens. Evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. Index these $H = 3$ templates by h . Let $e_h(c)$ be their unit-norm text embeddings and $d_{R,h}(c)$ the drops from Eq. 1. Define $n_c = \|H^{-1} \sum_h e_h(c)\|_2$ and the normalized aggregate $\bar{d}_c(R) = H^{-1} \sum_h d_{R,h}(c)/n_c$. We suppress R when fixed. A foil is a non-target class; $\mathcal{F}_t$ contains all such dataset classes. The held-out worst-foil margin is

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

This equals using normalized mean text embeddings; class-specific n_c can change rankings. Tables report raw target drop $H^{-1} \sum_h d_{R,h}(t)$, whereas the positive-target screen uses $\bar{d}_t$. Selection and evaluation use separate templates.

Let bbox precision be the fraction of selected patch centers inside target-class boxes. For CCI-top1, the fixed baseline

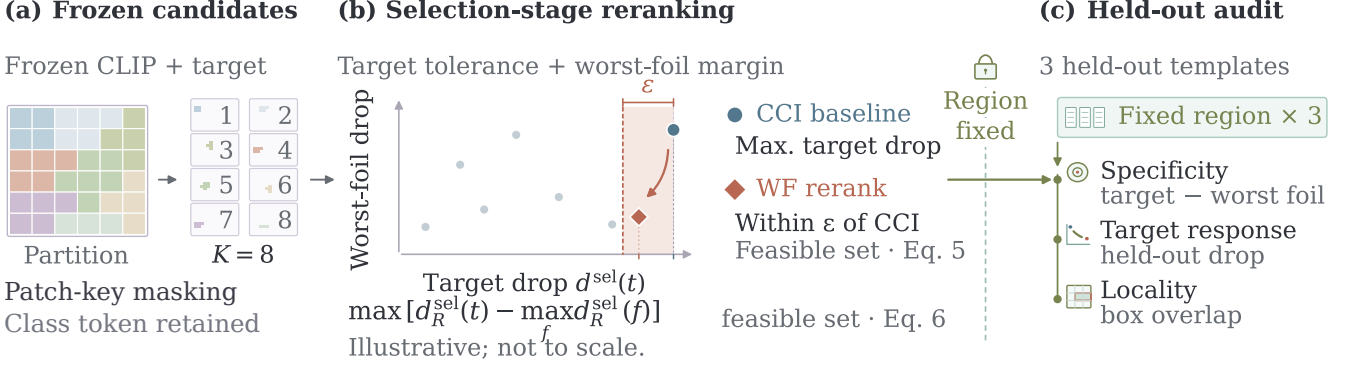


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. Schematic responses illustrate the selection rule.

screen-passers are $A = \{\text{bbox} \geq .5, \bar{d}_t - \text{mean}_f \bar{d}_f > 0\}$ and the failure set is $F = \{P < 0\}$. The mean is over $f \in \mathcal{F}_t$; A requires at least half of patch centers to overlap target boxes and a positive target-minus-mean-foil drop. To distinguish positive mean-foil contrast from positive target contribution, we also audit $A^+ = A \cap \{\bar{d}_t > 0\}$. The primary sign-repair population is $B = A \cap F$, fixed before reranking. Throughout, “failure” denotes $P < 0$, and “sign repair” denotes a transition from $P < 0$ to $P \geq 0$.

Target-constrained diagnostic probe. In Figure 1(b), M^{sel} measures selection-stage target-minus-worst-foil drop. The tolerance $\epsilon \geq 0$ defines the feasible subset $\mathcal{R}_\epsilon \subseteq \mathcal{R}(x)$:

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and worst-foil (WF) reranking

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

CCI is feasible by construction. We use $\epsilon = .02$ logit units and sweep $.01/.02/.05/.10/.20$ within each checkpoint.

Exact diagnostic oracles. For each image i , write $P_i(R) = P(R, t_i)$ for its target t_i . An oracle checks every candidate using held-out responses to count available repairs. With $\mathbf{1}\{\cdot\}$ the indicator (one if true, zero otherwise), the feasible sign-repair count on fixed baseline B is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1}\left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle searches all K candidates. A sign pass means $P_i(R) \geq 0$, without locality or positive-target constraints. Failures split into: no passing candidate (C_0); passing candidates but *none feasible* (C_1); a feasible pass missed by WF (C_2); and WF sign repair (r). Thus $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. We additionally count repairs

satisfying $J(R) = \{P_i(R) \geq 0, A(R), \bar{d}_t(R) > 0\}$; $A(R)$ applies the same full-foil screen to the new region, without changing fixed baseline B . Here r counts WF sign repairs and j counts those also satisfying J . Oracles measure available opportunity; WF selects using only its selection prompt.

Fixed-population foil controls. For image i , let $\mathcal{F}_i^{\text{abs}}$ remove all annotated-present categories from the non-target foils. The protocol holds R_{CCI} and full-foil screen A fixed; F_i^{abs} tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. Reducing the foil set weakly decreases failure, so we also match its cardinality. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils with $\bar{d}_f > \bar{d}_t$, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

With zero numerator when $k_i > N_i - m_i$, we compare $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{F_i^{\text{abs}}\} - q_i)$. Restricted sets are nonempty, and crowd annotations are excluded.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [11] and VOC2007 test [12] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (ViT-B with 16- or 32-pixel patches; 59,302 image–model records). The target is the largest instance’s category, using annotation area on COCO and box area on VOC. We require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means clusters L2-normalized projected patch features: $K = 8$, three initializations, 50 iterations, seed 1701 + i (sorted IDs shuffled with seed 1701 define i). Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax; MPS features are cast to float32. Target boxes are mapped by $224 / \min(W_{\text{img}}, H_{\text{img}})$ and center-crop offsets, where $W_{\text{img}}, H_{\text{img}}$ are the image width and height [13].

4 RESULTS AND ANALYSIS

Table 1: Fixed baseline A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.71]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.08	39.69	40.24	-0.55 [-1.21,+0.04]

Failure persists after annotation exclusion. The fixed screen A contains 17,726, 17,848, 1,209, and 1,227 images in table order. Exclusion lowers failure by only 1.14–1.39 percentage points of A ; 96.63–98.24% of fixed baseline failures B remain (Table 1). Table 1 separates this change from foil-count reduction: COCO exclusion has a larger effect than random removal of the same number of categories, whereas VOC Δ_A intervals include zero. The residual failure extends beyond the contribution of explicitly annotated co-occurring classes.

Positive-target and scene-count checks. The mismatch persists under positive-target filtering. Among A^+ , full-foil failure is 62.64–63.79% on COCO and 40.66–41.40% on VOC. After annotation exclusion, it remains 61.38–62.62% and 39.26–40.05%, respectively. Matched-random comparisons retain the COCO/VOC interval pattern. Even scenes with exactly two annotated categories have COCO annotation-absent failure of 63.39/64.95%.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11410	10697	695	0	18
COCO B/32	11562	10655	876	1	30
VOC B/16	511	371	135	0	5
VOC B/32	504	347	154	1	2

Candidate and budget bottlenecks differ by dataset. Table 2 shows that no candidate passes (C_0) for 92.16–93.75% of COCO screened failures B , versus 68.85–72.60% on VOC (Table 2). Unrestricted sign-repair capacity is thus 6.25–7.84% of COCO B and 27.40–31.15% of VOC B (Figure 2). The .02 budget excludes 96.43–98.09% of these opportunities, leaving feasible capacity of just 0.16–0.98% of B . WF misses at most one feasible opportunity per setting (C_2 in Table 2). The circle-to-diamond gaps in Figure 2 visualize the budget-excluded opportunities (C_1).

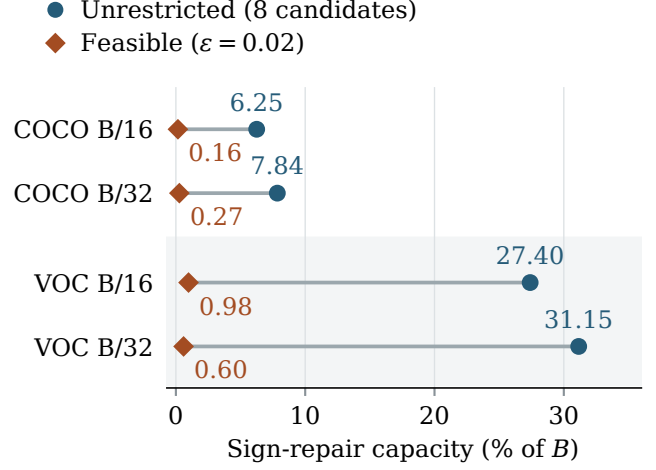


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2) and are computed by exact enumeration.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: percentage of all eligible images with multiple feasible candidates. O_B, r_B : oracle/actual sign-repair percentages of screened failures B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.95	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.60	0.40	-0.0007	+0.02
V32	0.20	34.64	7.14	6.75	-0.0161	+1.07

Budget relaxation helps in every setting. Table 3 reports oracle capacity $O_B = 100N_O/|B|$ and achieved repair $r_B = 100r/|B|$. The changes Δd_t and Δb are WF minus CCI averages over all eligible images, with bbox change expressed in percentage points. At .02, 95.21–95.60% of images have only one feasible candidate (Table 3); WF has little room to change the region. At .20, oracle sign-repair capacity rises to 1.57–2.05% of COCO B and 6.65–7.14% of VOC B . Raw-target changes become negative in all four settings, but bbox changes are negative on COCO and positive on VOC. Budget relaxation therefore has dataset-dependent locality effects. Each comparison uses the same checkpoint and fixed baseline population.

Table 4: Normalization sensitivity on fixed earlier-run A_{old} and earlier-run CCI regions. Raw/Norm: failure rates (%); Flip: percentage whose failure status differs. A_{old} denotes earlier-run screen-passers.

Setting	$ A_{\text{old}} $	Raw	Norm	Flip
COCO B/16	17,726	64.318	64.369	0.107
COCO B/32	17,847	64.762	64.778	0.073
VOC B/16	1,209	42.184	42.266	0.083
VOC B/32	1,227	41.157	41.157	0.000

Normalization has a small effect on the fixed audit. With earlier-run regions and A_{old} held fixed, raw and normalized failure rates differ by at most 0.083 percentage points; sign flips affect at most 0.107% of this population (Table 4). The hardest foil agrees on 99.42–99.84% of images. For example, COCO B/16 changes from 64.318% to 64.369%, while VOC B/32 stays at 41.157%. Table 4 therefore shows that normalization changes individual outcomes rarely and preserves the aggregate pattern.

Table 5: Actual sign repairs *within fixed baseline B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF–CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δbbox
COCO B/16	18	12	0.029	-19.53
			[-0.096,0.148]	[-38.12,-1.32]
COCO B/32	30	25	0.094	-14.04
			[0.036,0.148]	[-27.70,-1.74]
VOC B/16	5	3	-0.021	-16.87
			[-0.170,0.182]	[-26.11,-7.00]
VOC B/32	2	1	-0.078	-28.17
			[-0.115,-0.042]	[-78.57,22.22]

Which images are averaged? Table 3 averages over all eligible images, including unchanged regions. Table 5 averages only over WF sign repairs within fixed baseline B .

Target and locality changes after sign repair. Table 5 shows negative mean bbox changes in all four repair subsets: -14.04 to -28.17 percentage points. Target-score changes are positive on COCO and negative on VOC. Thus crossing the worst-foil threshold does not impose a common direction on target response or spatial overlap. In every setting, some sign repairs fail to retain the locality/mean-foil screen and positive target contribution ($j < r$ in Table 5). The joint feasible-oracle counts are 12/26/3/2. Outside the screened population, sign transitions can also regress: COCO B/32 has four opposite-sign regressions among all baseline passes. Counts over all failures or all switched regions have different denominators from the screened sign repairs reported here.

Table 6: Nonzero-threshold sensitivity on fixed baseline A . Entries are percentages of screen-passers A with $P < -\delta$, where $\delta \geq 0$ is the required failure magnitude in the normalized margin. Thresholds are evaluated post hoc in logit units.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.08	40.26	34.39	11.90

Failure persists at larger margin thresholds. At $\delta = .10$, 59.49–60.22% of COCO and 34.39–36.23% of VOC screen-passers A still fail (Table 6). Rates fall further at .50, so the magnitude and threshold matter. The threshold sweep characterizes the magnitude of failure within the normalized aggregate score.

What the comparisons establish. Table 1 isolates the effect of removing annotated-present classes from the automatic benefit of reducing the foil count. Table 2 and Figure 2 then separate candidate availability from budget feasibility: a better reranking rule alone can recover only the additional opportunities in C_2 at the fixed budget. Table 3 shows how enlarging that feasible set changes the attainable repair rate. Together, these comparisons locate the main constraints on sign repair in the candidate set and target-drop tolerance.

Records and uncertainty. The primary tables use one traceable release of candidate masks, selection responses, three-template held-out responses, and bbox records generated together. Checks verify patch partitions, bbox recomputation, response aggregation, candidate choices, and the failure decomposition. The frozen release includes candidate summaries, validation reports, configuration, and hashes; masks and full responses are retained locally. Table 4 retains earlier-run regions and population; three images changed selected regions between runs.

Confidence intervals (CIs) use 10,000 image bootstraps and percentile 2.5/97.5 limits, pointwise without multiplicity correction. Paired effects are differenced within image before resampling. Foil comparisons resample within fixed A or A^+ ; repair-cost intervals condition on actual repairs. Exact q_i is primary; 100 seeded draws check it.

5 CONCLUSION

Locality and positive mean-foil contrast leave substantial worst-foil failure in CCI-top1 regions, persisting after annotation exclusion and at larger margin thresholds. Candidate availability and target-drop constraints account for most limits on sign repair. Larger budgets increase capacity with setting-dependent target and locality effects. These findings support reporting worst-foil specificity alongside locality and mean-foil contrast when evaluating frozen CLIP region explanations.

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COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.